

Area under a polar curve

Key Points

Consider a polar curve of the form

$$r = f(\theta), \text{ for } \alpha \leq \theta \leq \beta$$

The area of the region bounded by the curve is given by

$$\int_{\alpha}^{\beta} \frac{1}{2} r^2 d\theta$$

Remember that r cannot be negative when $\alpha \leq \theta \leq \beta$ for this formula to work

Basic Skills

Q1. Find the area bounded by the following circles given in polar coordinates

- a) $r = 1$ b) $r = 2$ c) $r = 5$ d) $r = 10$

Q2. By using the formula given in the key points section, calculate the area of the region bounded by the polar curve $r = \sqrt{\theta}$, for $0 \leq \theta < 2\pi$.

Q3. Find the area bounded by the following polar curves

- a) $r = \theta$, where $0 \leq \theta \leq \pi$
b) $r = \theta^2$, where $0 \leq \theta \leq \pi$

Competency Skills

Q4. Can you verify that the area of a circle of radius R centred at the origin is indeed given by the formula πR^2

Q5. Find the area of the following polar curves

- a) $r = \sin(\theta)$, where $0 \leq \theta \leq \pi$
b) $r = \sin(\theta)$, where $0 \leq \theta < 2\pi$
c) Why are the areas of the curves in a) and b) the same?

Q6. What is the area of the region under the polar curve $r = 2 + \cos(\theta)$, where $0 \leq \theta < 2\pi$

Q7. What is the area of the region under the polar curve $r = \sin 3\theta$ where $-\pi < \theta \leq \pi$

Advanced Skills

Q8. What is the area of the region under the polar curve $\theta = \ln(r)$, where $1 \leq r \leq e^{2\pi}$

Q9. What is the area of the region enclosed by the line $y = \sqrt{3}x$, the line $y = \frac{1}{\sqrt{3}}x$ and the polar curve $r = 2 - 2\sin(\theta)$ for $0 \leq \theta < \pi$

Q10. What is the area of the region enclosed by the positive x -axis, the line $y = x$ and the curve given by the equation $y = x\sqrt{x^2 + y^2}$

Q11. Does the range of θ affect the value of the area under the curve for any polar curve $r(\theta)$

Extension

Q12. What is the area of the region bounded by the polar curve $r = \theta^{-\frac{1}{n}}$, where $0 \leq \theta \leq 2\pi$ and n is a positive integer bigger than 1. What happens as n gets bigger? Why is this?